Measuring and reporting performance

CSE Sem-4

System Performance

- Ideal performance of computer demands perfect match between machine capability and program behavior.
- Machine capability increased by better hardware technology, innovative architecture, efficient resource management.
- Program behavior difficult to predict due to its high dependence on application, algorithm design, data structure, programmer skill.
- As machine performance vary from program to program, achieving peak performance of machine, i.e. processor, is impossible.
- Some fundamental factors used to measure machine performance.

❖ **Instruction Count (I_c) :** Number of machine instructions to be executed in program. Different machine instructions required different number of clock cycles to execute.

❖ **Clock Rate and CPI:** $f = 1/\tau$ [τ = clock cycle time in ns,

f =frequency or clock rate in MHz]

$$\textbf{CPI (Cycles Per Instruction)} = \frac{\text{CPU Clock cycles for a program}}{\text{Instruction count } (Ic)}$$

❖ **CPU Time or execution time :** Time required to execute a program.

CPU clock cycles for a program $\times$ clock cycle time

= (CPI * Instruction count) $\times$ clock cycle time

CPU time (T) $= CPI \times I_c \times \tau$

- **MIPS Rate**: Million Instructions Per Second (Processor speed)

  C = Total no. of required clock cycles for execution

  $T = C \times \tau = C / f$

  Where, $CPI = C / Ic$,

  So, $C = CPI \times Ic$

  CPU time (T) $= CPI \times Ic \times \tau$

  $= (CPI \times Ic) / f$

  So, $Ic / T = f / CPI$

  MIPS $= Ic / (T \times 10^6)$ [T = Execution Time, Ic = Instruction Count]

  $= f / (CPI \times 10^6)$

  $= (f \times Ic) / (C \times 10^6)$

  MIPS = (Clock rate * Instruction count) / (Total clock cycle $*10^6$)

- MIPS is used for comparing the processor's performance based on similar architecture. Faster processors have a higher MIPS than slower computers.

❖ **MFLOPS:** Million Floating Point Operations Per Second

$$\text{MFLOPS} = \frac{\text{No of floating point operations in a program}}{\text{Execution Time} \times 10^6}$$

❖ **Throughput Rate or System throughput:** A system throughput is defined how many programs are executed in a system per unit time.

$$W_s = \text{(programs /second)}$$

❖ **CPU Throughput or CPU Performance:** Number of instructions executed per unit time. Performance is inversely proportional to execution time.

$$W_p = 1 / \text{CPU time} = f / (\text{CPI} \times I_c)$$
$$= (\text{MIPS} \times 10^6) / Ic$$

For multi-programmed system Ws < Wp, due to additional overheads of the system because of I/O, compiler and OS when multiple programs are interleaved for CPU execution in a time sharing operation.

- **Parallel Run Time, T(n) :** Time required to run a program on an n-processor computer.
  If n=1, T(1) denotes the run time of a program on a single processor, i.e. sequential operation.

- **Speed-up, S(n) :** Defined as the ratio of the time taken to run a program on a single processor T(1) to the time required to run it on a parallel computer with identical n-processors T(n).

$$S(n) = T(1) / T(n)$$

- **Efficiency (η):** Efficiency of a program on n processors is defined as the **ratio of speed-up achieved and the number of processors used to achieve it.**

$$\eta = E(n) = S(n) / n$$
$$= T(1) / [n * T(n)$$

Efficiency range between : $1/n \leq \eta \leq 1$ (Maximum efficiency of 1 never achieved due to parallel overheads)

❖ **Redundancy, R(n):** If O(n) represents the total number of unit operations performed by a n-processor system, then Redundancy in a parallel computation is defined as:

$$R(n) = \frac{O(n)}{O(1)} = \frac{\text{Unit Operations on n-processor system}}{\text{Unit Operations on 1-processor system}}$$

❖ **Utilization, U(n):** System utilization signifies the percentage of resources like processors, memories etc. that are busy during the execution of parallel program.

$$U(n) = R(n) * E(n)$$

❖ **Quality of Parallelism, Q(n):** Q(n) is directly proportional to speed-up and efficiency and inversely related to redundancy.

$$Q(n) = [\ S(n) * E(n)\ ] / R(n)$$

Scalability Metrics

Scalability is the **measure** of a **system's ability to increase or decrease in performance and cost** in response to changes in application and system processing demands.

- Machine size: The number of processors employed in a parallel computing system.
- Clock rate: Machine components driven by a clock rate which can scale up for better performance.
- Problem size: The amount of computational workload to solve a problem.
- Memory capacity: The amount of main memory used in a program execution.
- Communication overhead: For parallel processing, the amount of time spent for inter-processor communication, synchronization, remote memory access.

Numerical Question:

Consider two different machines, with two different instruction sets, both of which have a clock rate of 200MHz. The following measurements are recorded on the two machines running a given set of benchmark programs:

Instruction Type	Instruction Count (millions)	Cycles per Instruction
Instruction fetch and decode	8	1
Operand fetch	4	3
Execute	2	4
Store	4	3
Instruction fetch and decode	10	1
Operand fetch	8	2
Execute	2	4
Store	4	3

1. Determine CPI, MIPS rate, Execution Time for each machines
2. Compare the performance of two CPUs

Solution:

$$CPI_A = \frac{\sum CPI_i \times I_i}{I_c} = \frac{(8\times1+4\times3+2\times4+4\times3)\times10^6}{(8+4+2+4)\times10^6} \approx 2.22$$

$$MIPS_A = \frac{f}{CPI_A \times 10^6} = \frac{200\times10^6}{2.22\times10^6} = 90$$

$$CPU_A = \frac{I_c \times CPI_A}{f} = \frac{18\times10^6\times2.2}{200\times10^6} = 0.2 \text{ s}$$

$$CPI_B = \frac{\sum CPI_i \times I_i}{I_c} = \frac{(10\times1+8\times2+2\times4+4\times3)\times10^6}{(10+8+2+4)\times10^6} \approx 1.92$$

$$MIPS_B = \frac{f}{CPI_B \times 10^6} = \frac{200\times10^6}{1.92\times10^6} = 104$$

$$CPU_B = \frac{I_c \times CPI_B}{f} = \frac{24\times10^6\times1.92}{200\times10^6} = 0.23 \text{ s}$$

b. Even though, machine B has a higher MIPS than machine A, it needs a longer CPU time to execute the similar set of benchmark programs (instructions).